

Spectral Chronometry: Universal Clock Factorisation and the Higgs-Dilaton Test

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

10. Clock spectral fields

A local clock is represented not by a coordinate label but by a positive transition frequency $\omega_A(x)$. The invariant quantity is its phase accumulation

$$d\Phi_A = \omega_A ds_g.$$

Each individual clock defines a representative

$$g_{\mu\nu}^{(A)} = \left(\frac{\omega_A}{\omega_{A*}} \right)^2 g_{\mu\nu},$$

for which its intrinsic rate is constant. The nontrivial question is whether every clock defines the same representative.

11. Universal clock factorisation theorem

Theorem. Let $U \subset \mathcal{M}$ be connected and let $\{\omega_A\}$ be positive clock spectral fields. The following are equivalent:

1. There exists $\hat{g} \in [g]$ and positive constants $\bar{\omega}_A$ such that $\Phi_A[\gamma] = \bar{\omega}_A \int_\gamma ds_{\hat{g}}$ for every timelike curve γ in U .
2. Every pairwise ratio is constant: $d \ln(\omega_A/\omega_B) = 0$.
3. There is one positive Weyl-weight -1 scalar χ and constants c_A such that $\omega_A = c_A \chi$.

Proof. Write $\hat{g} = e^{2\sigma}g$. Equality of phase integrands gives $\omega_A = \bar{\omega}_A e^\sigma$, proving constant ratios. If all ratios are constant, choose one reference clock and set χ proportional to its spectrum; the other spectra are constant multiples. Conversely, if $\omega_A = c_A \chi$, then $\hat{g} = (\chi/\chi_*)^2 g$ gives

$$\Phi_A = c_A \chi_* \int ds_{\hat{g}}.$$

Uniqueness is up to one global multiplicative unit convention. $\square$

The theorem is elementary once stated. Its value is that it turns the phrase “all clocks measure the same proper time” into an exact integrability condition.

12. Chronometric shear and clock-space rank

Define

$$\boxed{\mathcal{S}_{AB} = d \ln \frac{\omega_A}{\omega_B}.$$

This is invariant under Weyl rescaling. A useful clock-space decomposition begins with

$$\ell_A = \ln \omega_A,$$

where conformal gauge shifts all components in the common direction $(1, \dots, 1)$. Physical clock disagreement lives in the quotient of log-spectrum space by that common direction.

With positive weights w_A satisfying $\sum_A w_A = 1$, define

$$B_\mu^A = -\partial_\mu \ln \omega_A, \quad W_\mu = \sum_A w_A B_\mu^A,$$

$$\Sigma_\mu^A = B_\mu^A - W_\mu.$$

Then Σ^A and $\mathcal{S}_{AB}$ are gauge invariant, and

$$\mathfrak{C}_{\mu\nu} = \sum_A w_A \Sigma_\mu^A \Sigma_\nu^A$$

is positive semidefinite. It vanishes exactly when one chronometric metric exists.

Suppose spectra depend on r nonuniversal dimensionless fields q^I :

$$\delta \ln \omega_A = \delta \ln \chi + K_{AI} \delta q^I.$$

Clock ratios obey

$$\delta \ln \frac{\omega_A}{\omega_B} = (K_{AI} - K_{BI}) \delta q^I,$$

so

$$\boxed{\text{rank} [\delta \ln(\omega_A/\omega_B)] \leq r.}$$

One nonuniversal scalar predicts a rank-one network across arbitrarily many clocks. Exact universal chronometry is rank zero.

13. The role of electromagnetism

Source-free Maxwell theory in four dimensions is conformally invariant. It transports phase, null structure, polarisation, and causal information, but it does not select an absolute spectral scale. A free wave's frequency is state data.

Matter interactions create bound-state gaps. A hydrogenic scale is schematically

$$E_{\text{Ry}} \sim m_e \alpha_{\text{em}}^2, \quad m_e = \frac{y_e \hbar}{\sqrt{2}}.$$

If $E_A = c_A \chi$, then

$$d\Phi_A = \frac{E_A}{\hbar} ds_g = c_A \chi_* ds_{\hat{g}}.$$

The division of labour is therefore:

χ : universal normalisation, Higgs and QCD : transmission into material spectra, electromagnetism : phase organisation, transport, and readout.
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The correct statement is not “the Higgs and photons create time.” It is that a deeper scale field calibrates duration, material sectors relay the scale, and electromagnetic phase makes the count operational.

14. Why the Standard Model Higgs is insufficient by itself

A general clock frequency has the structure

$$\omega_A = h F_A \left(\alpha_{\text{em}}, g_s, y_f, \frac{\Lambda_{\text{QCD}}}{h}, \frac{M_{\text{P}}}{h}, \dots \right).$$

The ordinary Higgs sets elementary electroweak masses, but it does not impose

$$d \ln \frac{\Lambda_{\text{QCD}}}{h} = 0, \quad d \ln \frac{M_{\text{P}}}{h} = 0.$$

Nuclear clocks depend on confinement, quark masses, nuclear magnetic moments, and many-body structure. The Higgs is a relay, not automatically a universal source.

Higgs-only induced gravity would require

$$M_{\text{P}}^2 = \xi_h v^2,$$

so

$$\xi_h \sim 10^{32},$$

far beyond the minimal phenomenologically plausible range [Atkins2012]. Using the sole radial Higgs mode as a local Weyl compensator also removes the physical scalar remaining after the electroweak Goldstones are eaten. A second radial direction is structurally required.

15. Higgs-dilaton and deeper-scale completion

Introduce a positive scalar χ and impose

$h = \alpha \chi, \quad \Lambda_{\text{QCD}} = c_Q \chi, \quad M_{\text{P}} = c_P \chi.$
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If all dimensionless couplings are fixed, every physical gap is

$$\omega_A = c_A \chi.$$

Write

$$h = \rho \sin \theta, \quad \chi = \rho \cos \theta.$$

The radial coordinate ρ is the common scale; the angular coordinate θ measures Higgs/dilaton alignment. A pure radial fluctuation rescales every clock together and is invisible to local ratios. An angular fluctuation produces

$$\delta \ln \frac{\omega_A}{\omega_B} = (K_A - K_B) \delta \theta.$$

A healthy globally scale-invariant representative has

$$F(h, \chi) = \xi_h h^2 + \xi_\chi \chi^2,$$

$$V(h, \chi) = \frac{\lambda}{4} (h^2 - \alpha^2 \chi^2)^2 + \beta \chi^4.$$

In the Einstein frame, the scalar field-space metric

$$G_{ij} = \frac{M_{\text{P}}^2}{F} \delta_{ij} + \frac{3M_{\text{P}}^2}{2F^2} F_{,i} F_{,j}$$

is positive for positive Jordan kinetic signs and $F > 0$. In the exact global scale limit, one radial dilaton direction can remain flat; local Weyl symmetry can gauge it; quantum breaking can lift it. These are different physical theories and should not be conflated [Shaposhnikov2009; Foot2007].